Donghao Wu

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EDUCATION

George Washington University, Washington, DC — Expected Jan 2027
Ph.D. in Economics
Dissertation Committee: Leah Brooks (Chair), Paul Carrillo, and Alessandra Fenizia

Georgetown University, Washington, DC — May 2020
M.S. in Economics

University of Maryland, College Park, MD — May 2018
B.A. in Economics

RESEARCH FIELDS

Primary: Urban Economics, Applied Microeconomics
Secondary: Public Policy, Monetary Economics

RESEARCH PAPERS

How Does City-Center Spending Respond to COVID-19 Relative to Surrounding Areas? Evidence from U.S. Metropolitan Areas

[Job Market Paper]

- Uses card-transaction data aggregated to fine geographic cells and a difference-in-differences and event-study design to compare CBD and non-CBD spending across 352 U.S. metropolitan areas before and after March 2020.
- Finds that CBD spending fell by approximately 42% relative to non-CBD spending through 2020 and that the gap remained visible through 2024; higher-income consumers experienced an additional 6.3% relative decline.
- Relates city-specific estimates to pre-pandemic population and commuting patterns; declines are larger in more populous metropolitan areas and those with more non-CBD-to-CBD commuting.

Temporary Tax Relief and Consumer Demand: Evidence from US Sales Tax Holidays

(with Rui Zhong) [Working Paper]

- Studies 2023 back-to-school sales tax holidays using daily card transactions aggregated to small geographic cells and a difference-in-differences design with geographic-cell and date fixed effects.
- Finds that computer spending rises by approximately $547 per hexagon-day during the first three days of a holiday, with little evidence of increased spending on clothing or school supplies.
- Documents spatially uneven responses within participating states and no clear offsetting losses in nearby comparison-state locations, suggesting limited cross-border displacement.

Monetary Policy Transmission to Consumption: Evidence from Transaction-Level Data

(with Ammu Lavanya) [Work in Progress]

- Uses U.S. card transactions from 2019 to 2024, mapped to 15 Personal Consumption Expenditure categories, and local projections with high-frequency monetary policy shocks.
- Preliminary estimates show that positive contractionary shocks reduce spending and negative expansionary shocks raise it; ongoing work examines asymmetry and heterogeneity across income groups and consumption categories.

RESEARCH EXPERIENCE

Federal Statistical Research Data Center, Washington, DC — May 2023 - August 2023
Research Assistant to Leah Brooks (George Washington University) and Rachel Meltzer (Harvard University)

- Built analysis-ready research files from restricted establishment-level Census microdata and proprietary CoStar and InfoUSA data to study retail-space supply, demand, and vacancy across five major U.S. cities.
- Cleaned, standardized, and linked large datasets in R across locations, establishments, leases, and time; analyzed cross-city patterns in leasing activity, rents, and space utilization.

George Washington University, Department of Economics — June 2021 - August 2021
Teaching and Research Assistant to Irene Foster

- Developed guided notes and module quizzes for the mathematics textbook “I Am Not a Math Person!” for use by more than 300 undergraduate economics students.
- Researched economic-data applications for assessment questions and proofread the manuscript for classroom adoption.

TEACHING EXPERIENCE

John Locke Institute, Public Policy Symposium, Princeton, NJ August 2026

Preceptor

- Led daily precepts in which students evaluated competing policy arguments, developed evidence-based positions, and connected lectures to contemporary questions; guided discussion with targeted prompts while preserving student ownership of the debate.
- Designed and facilitated interactive morning activities that introduced each day's policy theme, including public choice, taxation, budget deficits, immigration, and education funding.
- Organized and supported extracurricular simulations, including Model United Nations-style and cabinet-level crisis exercises; managed briefings, procedural motions, moderated debate, and voting.
- Mentored student teams as they developed public-policy briefs and presentations, offering feedback on tradeoffs, implementation, and the clarity of their recommendations.

George Washington University, Department of Economics August 2021 - May 2026

Graduate Teaching Assistant to Kenneth Danger and Anthony Kassekert

- Served as Head Teaching Assistant for the M.S. in Economics program (Spring 2023 - May 2026), coordinating TA assignments and schedules across courses.
- Led office hours, discussion sections, homework workshops, and exam reviews for Machine Learning, Mathematics for Economics and Probability, and Statistics for Economics.
- Developed questions for mathematics midterms and final examinations; graded assignments and exams and provided feedback to strengthen students' quantitative reasoning.
- Nominated for the 2024 Philip Amsterdam Graduate Student Teaching Award.

University of Maryland, Department of Economics, College Park, MD August 2018 - December 2018

Teaching Assistant to Peter Coughlin - ECON 451: Public Choice

- Managed and updated the course website using HTML, held weekly office hours, and led review sessions before midterm and final examinations.

PROFESSIONAL EXPERIENCE

Intelliimpact, Remote February 2026 - April 2026

Data Science Intern

- Built analytics pipelines integrating AMI time-series, weather, and grid-connectivity data and engineered features from load patterns, seasonality, and lagged demand for U.S. electric utility clients.
- Developed and evaluated tree-based forecasting and outage-risk models, improving predictive accuracy by approximately 10-20%, and applied clustering and anomaly-detection methods to support operational monitoring.

Public Spend Forum, Washington, DC February 2019 - May 2020

Research Analyst Intern

- Analyzed procurement-platform data, FPDS, USAspending, and Crunchbase and designed SME surveys to identify barriers to government contracting and inform supplier onboarding.
- Classified more than 100 emerging-technology categories, developed Tableau dashboards, and expanded the supplier database by onboarding more than 43,000 technology-focused SMEs.

METHODS AND TECHNICAL SKILLS

Econometrics: Difference-in-differences, event studies, local projections, instrumental variables, A/B testing

Programming: Python (pandas, NumPy, scikit-learn), R, Stata, SQL

Data: Large-scale transaction data, foot-traffic data, and geospatial data

Visualization: Tableau, Matplotlib, Seaborn